Probability Theory: Fundamentals and Applications

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Probability Theory: Fundamentals and Applications

1. Introduction

Probability theory provides a mathematical framework for quantifying uncertainty. In mathematics, **probability is a measure of uncertainty of events in a random experiment**. It quantifies the likelihood that a particular outcome will occur, using numbers between 0 (impossible) and 1 (certain).

The axiomatic approach to probability, formulated by A.N. Kolmogorov, treats probability as a function of outcomes of an experiment and establishes the foundation for modern probability theory.

2. Basic Concepts and Definitions

Sample Space and Events

- **Sample Space (S)**: The set of all possible outcomes of a random experiment
- **Event**: A subset of the sample space
- **Outcome**: A single result of a random experiment

Random Variables

A **random variable** is a real-valued function whose domain is the sample space of a random experiment. It assigns numerical values to the outcomes of a random experiment.

In probability theory, we're often interested not just in particular outcomes, but in numerical values associated with those outcomes. For example: - When tossing two dice, we might be interested in the sum of the numbers - When tossing a coin multiple times, we might want to count the number of heads

3. Basic Probability Rules

Fundamental Rules

1. **Addition Rule**: For mutually exclusive events A and B:

```
KaTeX error: node:internal/modules/cjs/loader:1424
  throw err;
  ^

Error: Cannot find module 'katex'
```

```
Require stack:
- /home2/ameyk/lma-project/agents/content_generation/_katex_node/k
    at Module._resolveFilename (node:internal/modules/cjs/loader:1
    at defaultResolveImpl (node:internal/modules/cjs/loader:1059:1
    at resolveForCJSWithHooks (node:internal/modules/cjs/loader:10
    at Module._load (node:internal/modules/cjs/loader:1227:37)
    at TracingChannel.traceSync (node:diagnostics_channel:328:14)
    at wrapModuleLoad (node:internal/modules/cjs/loader:245:24)
    at Module.require (node:internal/modules/cjs/loader:1504:12)
    at require (node:internal/modules/helpers:152:16)
    at Object. (/home2/ameyk/lma-project/agents/content_generation
    at Module._compile (node:internal/modules/cjs/loader:1761:14)
  code: 'MODULE_NOT_FOUND',
  requireStack: [
    '/home2/ameyk/lma-project/agents/content_generation/_katex_noo
  ]
}

Node.js v24.11.1
```

2. **Multiplication Rule**: For independent events A and B:

```
KaTeX error: node:internal/modules/cjs/loader:1424
  throw err;
  ^

Error: Cannot find module 'katex'
Require stack:
- /home2/ameyk/lma-project/agents/content_generation/_katex_node/k
    at Module._resolveFilename (node:internal/modules/cjs/loader:1
    at defaultResolveImpl (node:internal/modules/cjs/loader:1059:1
    at resolveForCJSWithHooks (node:internal/modules/cjs/loader:10
    at Module._load (node:internal/modules/cjs/loader:1227:37)
    at TracingChannel.traceSync (node:diagnostics_channel:328:14)
    at wrapModuleLoad (node:internal/modules/cjs/loader:245:24)
    at Module.require (node:internal/modules/cjs/loader:1504:12)
    at require (node:internal/modules/helpers:152:16)
```

```
    at Object. (/home2/ameyk/lma-project/agents/content_generation
    at Module._compile (node:internal/modules/cjs/loader:1761:14)
  code: 'MODULE_NOT_FOUND',
  requireStack: [
    '/home2/ameyk/lma-project/agents/content_generation/_katex_nod
  ]
}

Node.js v24.11.1
```

3. **Complement Rule**:

```
KaTeX error: node:internal/modules/cjs/loader:1424
  throw err;
  ^

Error: Cannot find module 'katex'
Require stack:
- /home2/ameyk/lma-project/agents/content_generation/_katex_node/k
    at Module._resolveFilename (node:internal/modules/cjs/loader:1
    at defaultResolveImpl (node:internal/modules/cjs/loader:1059:1
    at resolveForCJSWithHooks (node:internal/modules/cjs/loader:10
    at Module._load (node:internal/modules/cjs/loader:1227:37)
    at TracingChannel.traceSync (node:diagnostics_channel:328:14)
    at wrapModuleLoad (node:internal/modules/cjs/loader:245:24)
    at Module.require (node:internal/modules/cjs/loader:1504:12)
    at require (node:internal/modules/helpers:152:16)
    at Object. (/home2/ameyk/lma-project/agents/content_generation
    at Module._compile (node:internal/modules/cjs/loader:1761:14)
  code: 'MODULE_NOT_FOUND',
  requireStack: [
    '/home2/ameyk/lma-project/agents/content_generation/_katex_nod
  ]
}

Node.js v24.11.1
```

Properties of Probability

- All probabilities are between 0 and 1: `KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1`
- The probability of the entire sample space is 1: `KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/`

helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1

- The probability of an impossible event is 0: KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1

4. Examples with Dice and Coins

Coin Tossing Examples

Single Coin Toss

For a fair coin: - `KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1`

Two Coin Tosses

Consider tossing a coin two times in succession. The sample space is:

```
KaTeX error: node:internal/modules/cjs/loader:1424
  throw err;
  ^
```

```
Error: Cannot find module 'katex'
Require stack:
- /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex
    at Module._resolveFilename (node:internal/modules/cjs/loader:1421:
    at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19)
    at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:2
    at Module._load (node:internal/modules/cjs/loader:1227:37)
    at TracingChannel.traceSync (node:diagnostics_channel:328:14)
    at wrapModuleLoad (node:internal/modules/cjs/loader:245:24)
    at Module.require (node:internal/modules/cjs/loader:1504:12)
    at require (node:internal/modules/helpers:152:16)
    at Object. (/home2/ameyk/lma-project/agents/content_generation/_ka
    at Module._compile (node:internal/modules/cjs/loader:1761:14) {
  code: 'MODULE_NOT_FOUND',
  requireStack: [
    '/home2/ameyk/lma-project/agents/content_generation/_katex_node/ka
  ]
}

Node.js v24.11.1
```

Example 1: Let X denote the number of heads obtained: - `KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/`

`modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1 - KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1 - KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND',`

`requireStack: [ '/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js' ] } Node.js v24.11.1` - `KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1`

Example 2: Let Y denote the number of heads minus the number of tails: - `KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/`

`modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1 - KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1 - KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND',`

requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1 - KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1

Multiple Coin Tosses

- **Three Coins**: KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/

agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1

- **Four Coins**: KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1

Dice Examples

Single Die

For a fair six-sided die, each outcome has probability KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl

(node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] }
Node.js v24.11.1 :- KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] }
Node.js v24.11.1 - KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/

`loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1`

Two Dice

When rolling two fair dice, there are 36 equally likely outcomes: - `KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1` (outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1)) - `KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /`

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5. Conditional Probability

Definition

Conditional Probability answers the question: "What's the probability of event E happening, given that event F has already occurred?"

If E and F are two events associated with the same sample space of a random experiment, the conditional probability of event E given that F has occurred is:

```
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  throw err;
  ^

Error: Cannot find module 'katex'
Require stack:
- /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex
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    at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:2
    at Module._load (node:internal/modules/cjs/loader:1227:37)
    at TracingChannel.traceSync (node:diagnostics_channel:328:14)
    at wrapModuleLoad (node:internal/modules/cjs/loader:245:24)
    at Module.require (node:internal/modules/cjs/loader:1504:12)
    at require (node:internal/modules/helpers:152:16)
    at Object. (/home2/ameyk/lma-project/agents/content_generation/_ka
    at Module._compile (node:internal/modules/cjs/loader:1761:14) {
  code: 'MODULE_NOT_FOUND',
  requireStack: [
    '/home2/ameyk/lma-project/agents/content_generation/_katex_node/ka
  ]
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Node.js v24.11.1
```

provided `KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1`

Understanding Conditional Probability

Conditional probability is crucial for understanding dependent events and forms the basis for more advanced topics like Bayes' theorem. It represents updated probability based on new information.

Applications of Conditional Probability

1. **Drawing Cards Without Replacement**: The probability of drawing a second ace depends on whether the first card drawn was an ace
2. **Medical Diagnosis Testing**: The probability of having a disease given a positive test result
3. **Quality Control**: The probability of a defective item given it comes from a particular machine

Example: Card Drawing

Consider drawing two cards from a standard deck without replacement: - `KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/`

content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1 - KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad (node:internal/modules/cjs/loader:245:24) at Module.require (node:internal/modules/cjs/loader:1504:12) at require (node:internal/modules/helpers:152:16) at Object. (/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js:3:15) at Module._compile (node:internal/modules/cjs/loader:1761:14) { code: 'MODULE_NOT_FOUND', requireStack: ['/home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js'] } Node.js v24.11.1

6. Practice Problems

Problem 1: Dice Rolling

What is the probability of rolling a sum of 8 with two fair dice?

Solution: The favorable outcomes are (2,6), (3,5), (4,4), (5,3), (6,2).

```
KaTeX error: node:internal/modules/cjs/loader:1424
  throw err;
  ^

Error: Cannot find module 'katex'
Require stack:
- /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex
```

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    at Module._resolveFilename (node:internal/modules/cjs/loader:1421:
    at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19)
    at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:2
    at Module._load (node:internal/modules/cjs/loader:1227:37)
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    at Module._compile (node:internal/modules/cjs/loader:1761:14) {
  code: 'MODULE_NOT_FOUND',
  requireStack: [
    '/home2/ameyk/lma-project/agents/content_generation/_katex_node/ka
  ]
}

Node.js v24.11.1
```

Problem 2: Conditional Probability with Coins

If you toss three coins and observe at least one head, what's the probability that all three are heads?

Solution: - Event A: All three heads = {HHH} - Event B: At least one head = {HHH, HHT, HTH, HTT, THH, THT, TTH} - `KaTeX error: node:internal/modules/cjs/loader:1424 throw err; ^ Error: Cannot find module 'katex' Require stack: - /home2/ameyk/lma-project/agents/content_generation/_katex_node/katex_renderer.js at Module._resolveFilename (node:internal/modules/cjs/loader:1421:15) at defaultResolveImpl (node:internal/modules/cjs/loader:1059:19) at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1064:22) at Module._load (node:internal/modules/cjs/loader:1227:37) at TracingChannel.traceSync (node:diagnostics_channel:328:14) at wrapModuleLoad`

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Node.js v24.11.1

7. Real-World Applications

Probability theory has numerous practical applications:

- **Weather Forecasting**: Predicting the likelihood of rain or storms
- **Insurance**: Calculating premiums based on risk assessment
- **Finance**: Portfolio optimization and risk management
- **Medicine**: Clinical trial analysis and diagnostic testing
- **Engineering**: Reliability analysis and quality control
- **Sports**: Performance analysis and outcome prediction

8. Conclusion

Probability theory provides essential tools for understanding and quantifying uncertainty in various fields. The fundamental concepts covered—basic probability rules, random variables, and conditional probability—form the foundation for more advanced statistical analysis and decision-making under uncertainty.

Understanding these concepts enables us to make informed decisions when faced with uncertain outcomes and provides a mathematical framework for analyzing random phenomena in science, engineering, and everyday life.

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